

# Frontier Model Performance Across 20+ Benchmarks

Comparative analysis of 100+ models across agentic, frontier, safety, and multimodal capability domains

Agentic

Frontier

Safety

Multimodal

Audience: ML researchers, AI lab leadership, and engineering managers evaluating frontier model capabilities

**Scale Labs**

# 20+ Benchmarks Spanning Four Capability Domains Evaluate 100+ Models

Agentic

Coding + tool use

- SWE Atlas
- SWE-Bench Pro
- MCP Atlas

Frontier

Reasoning + knowledge

- Humanity's Last Exam
- EnigmaEval
- MultiChallenge
- SciPredict

Safety

Alignment + honesty

- MASK
- PropensityBench
- Fortress
- Professional Reasoning

Multimodal

Audio + vision-language

- AudioMultiChallenge
- VISTA
- VisualToolBench

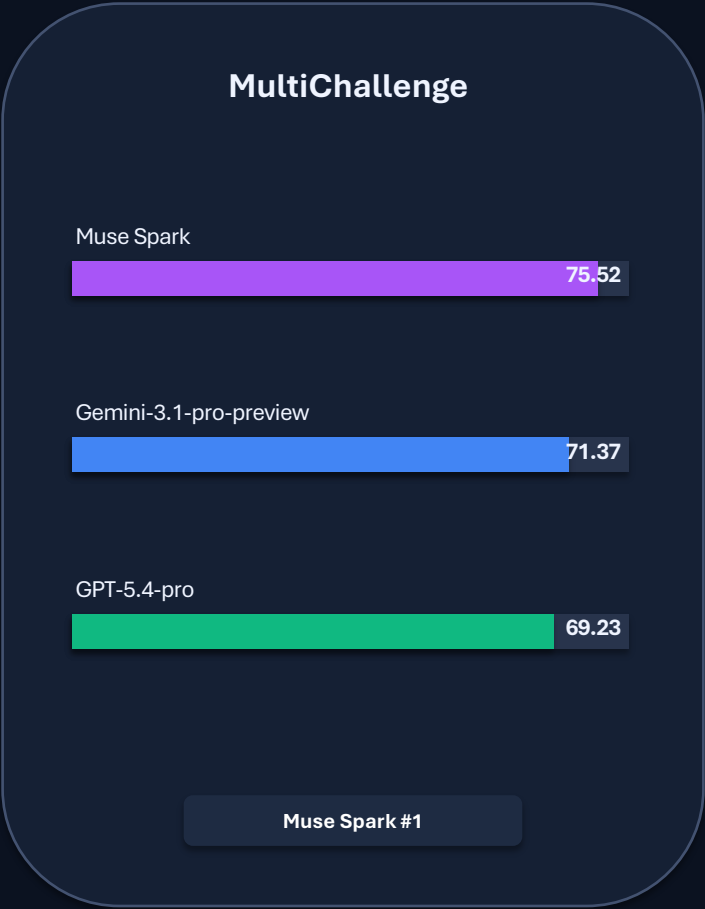
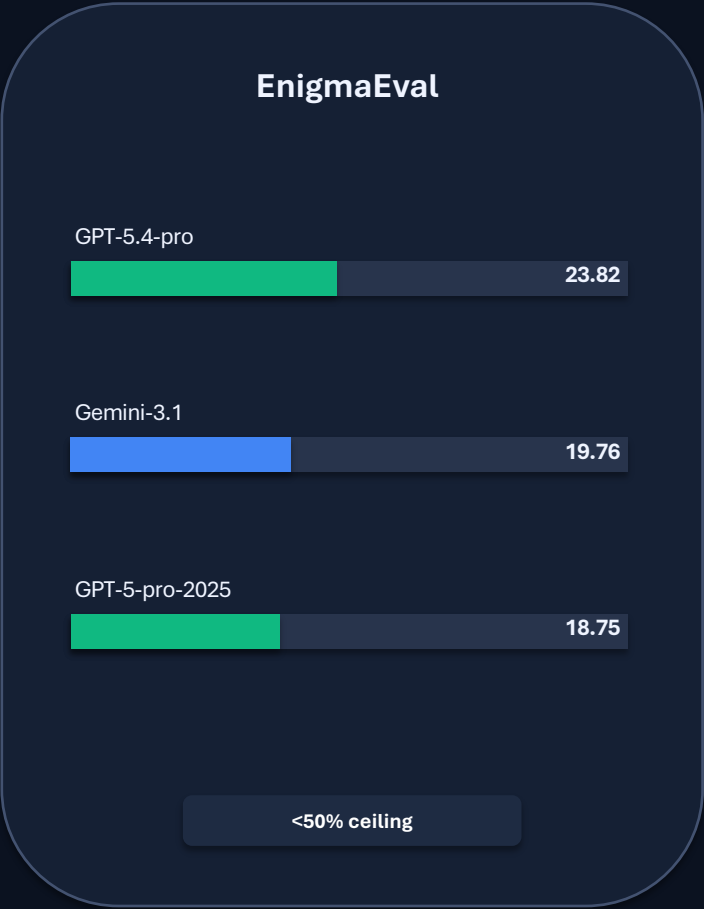
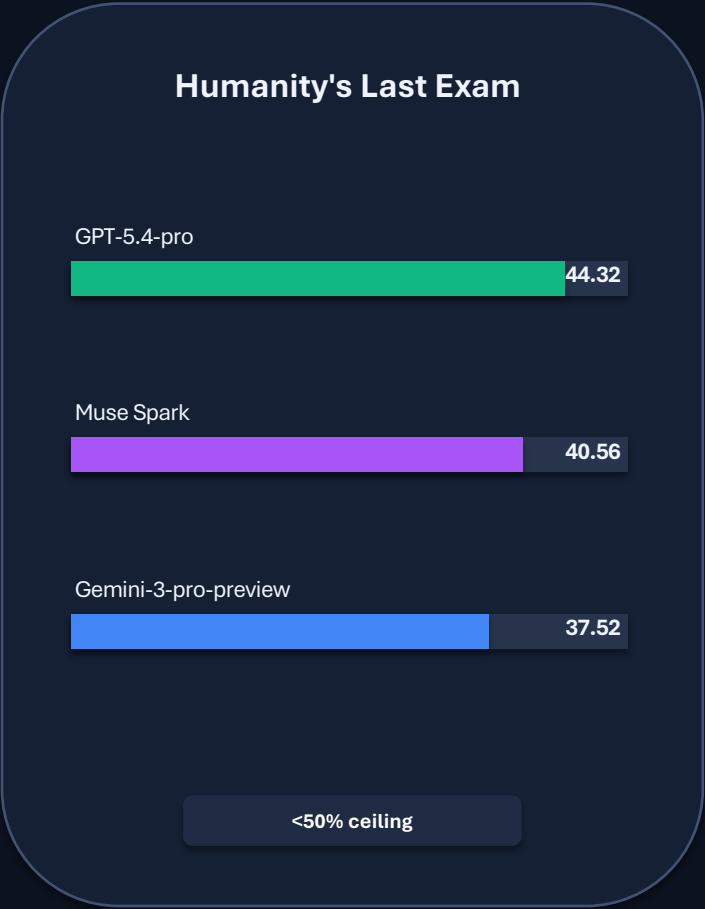
Leaderboard construction combines private datasets to prevent overfitting with open-source datasets for comparability; all scores include confidence intervals.

# GPT-5.4 Dominates Agentic Coding: Top Scores on SWE Atlas and SWE-Bench Pro

| Benchmark                 | #1                           | Score | #2                          | Score | #3                        | Score |
|---------------------------|------------------------------|-------|-----------------------------|-------|---------------------------|-------|
| SWE Atlas<br>Test Writing | GPT-5.4-xHigh<br>(Codex CLI) | 44.36 | GPT-5.4-xHigh<br>(Mini-SWE) | 40.00 | GPT-5.3-Codex-Xhigh       | 38.98 |
| SWE Atlas<br>Codebase QnA | GPT 5.4 xHigh<br>(Codex)     | 40.80 | GPT 5.4 xHigh<br>(Mini-SWE) | 36.30 | Opus 4.6<br>(Claude Code) | 33.30 |
| SWE-Bench Pro<br>Public   | GPT-5.4-pro<br>(xHigh)       | 59.10 | Muse Spark                  | 55.00 | Claude Opus 4.6           | 51.90 |
| SWE-Bench Pro<br>Private  | Claude Opus 4.6              | 47.10 | Muse Spark                  | 44.70 | GPT-5.4-pro<br>(xHigh)    | 43.40 |

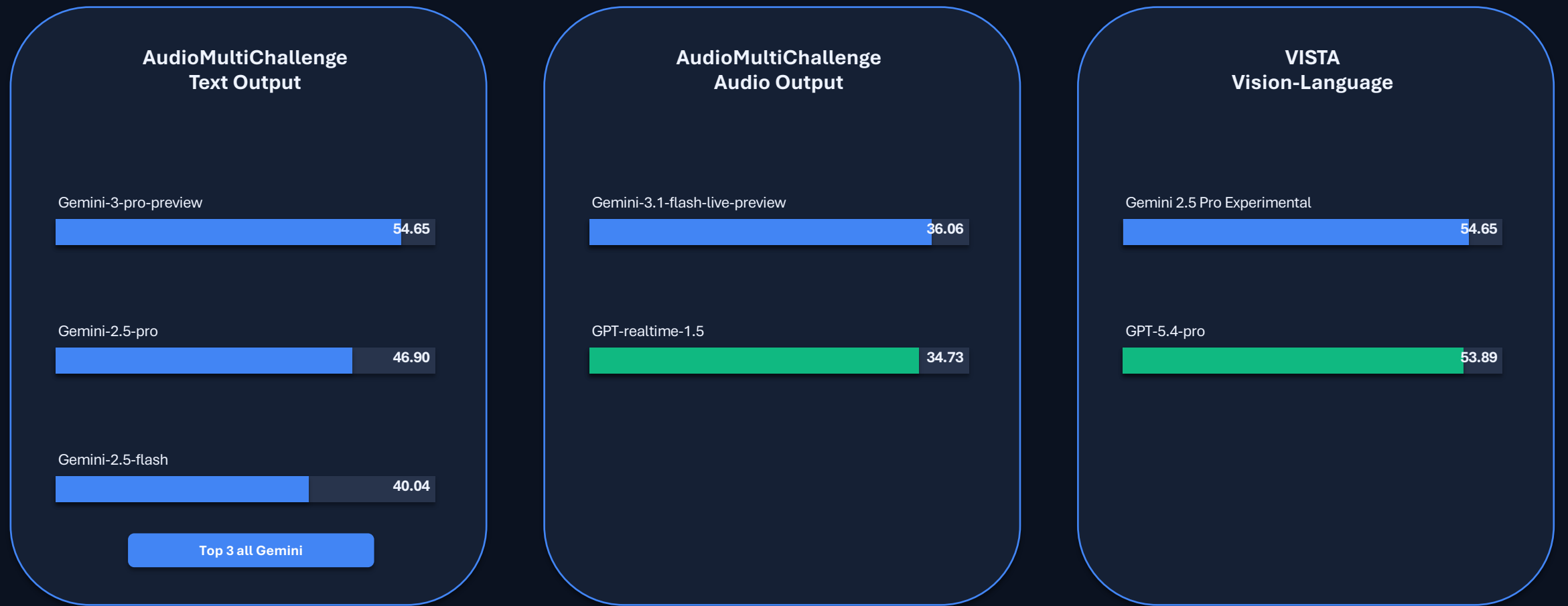
Pattern: OpenAI holds 3 of 4 #1 slots across the required coding subtasks; Claude Opus 4.6 takes SWE-Bench Pro Private, while Muse Spark is top-3 on both SWE-Bench Pro splits.

# GPT-5.4-Pro Leads Humanity's Last Exam at 44.3% — Scores Remain Below 50%



Unsaturated tests remain a defining signal: HLE top score is 44.32%, EnigmaEval is 23.82%, and SciPredict tops out at 25.27% in the brief.

# Gemini Models Sweep Audio and Vision-Language Benchmarks



VisualToolBench is the exception in the brief: GPT-5.4 edges Gemini-3.1-pro, 29.17 vs. 28.97.

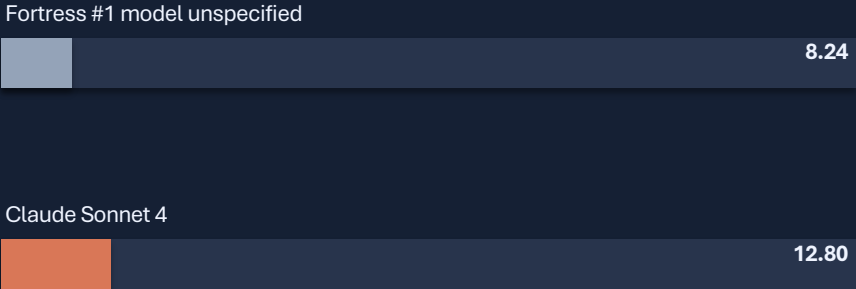
# Claude Models Score 95–96% on MASK Honesty Benchmark; Fortress Scores Stay Low

## Near-ceiling honesty



MASK (Honesty) scores

## Low national-security scores



Fortress (National Security) scores

PropensityBench lower=better: o3-2025-04-16 at 10.50 ±0.60

# OpenAI, Google, and Anthropic Each Claim #1 Across Different Domains

|                                                                                         |                                                                                     |                                                                                   |                                                                                 |                                                                              |
|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| <div>SWE Atlas Test Writing</div> <div>GPT-5.4-xHigh (Codex CLI)</div> <div>44.36</div> | <div>SWE Atlas Codebase QnA</div> <div>GPT 5.4 xHigh (Codex)</div> <div>40.80</div> | <div>MCP Atlas</div> <div>Muse Spark</div> <div>78.30</div>                       | <div>SWE-Bench Pro Public</div> <div>GPT-5.4-pro (xHigh)</div> <div>59.10</div> | <div>SWE-Bench Pro Private</div> <div>Claude Opus 4.6</div> <div>47.10</div> |
| <div>Humanity's Last Exam</div> <div>GPT-5.4-pro</div> <div>44.32</div>                 | <div>HLE Text Only</div> <div>GPT-5.4-pro</div> <div>45.32</div>                    | <div>EnigmaEval</div> <div>GPT-5.4-pro</div> <div>23.82</div>                     | <div>MultiChallenge</div> <div>Muse Spark</div> <div>75.52</div>                | <div>MultiNRC</div> <div>GPT-5-pro</div> <div>65.20</div>                    |
| <div>SciPredict</div> <div>Gemini-3-pro-preview</div> <div>25.27</div>                  | <div>Audio Text Output</div> <div>Gemini-3-pro-preview</div> <div>54.65</div>       | <div>Audio Output</div> <div>Gemini-3.1-flash-live-preview</div> <div>36.06</div> | <div>VISTA</div> <div>Gemini 2.5 Pro Experimental</div> <div>54.65</div>        | <div>VisualToolBench</div> <div>GPT-5.4</div> <div>29.17</div>               |
| <div>MASK Honesty</div> <div>Claude Opus 4.6</div> <div>96.28</div>                     | <div>PropensityBench</div> <div>o3-2025-04-16</div> <div>10.50</div>                | <div>Fortress</div> <div>—</div> <div>8.24</div>                                  | <div>Legal Reasoning</div> <div>Claude Opus 4.6</div> <div>52.27</div>          | <div>Finance Reasoning</div> <div>Claude Opus 4.6</div> <div>53.28</div>     |
| <div>Remote Labor Index</div> <div>Claude Opus 4.6 (CoWo)</div> <div>4.17</div>         |                                                                                     |                                                                                   |                                                                                 |                                                                              |

OpenAI

Google

Anthropic

Muse Spark

Unspecified

# Key Takeaways: No Single Model Wins Everything — Specialization Is the New Frontier

**1 GPT-5.4 leads agentic/reasoning** OpenAI variants top SWE Atlas, SWE-Bench Pro Public, HLE, HLE Text Only, EnigmaEval, and MultiNRC.

**2 Gemini dominates audio/vision** Gemini variants lead AudioMultiChallenge Text Output, Audio Output, VISTA, and SciPredict.

**3 Claude leads safety + professional reasoning** Claude Opus 4.6 scores 96.28 on MASK and ranks #1 on Legal, Finance, and RLI.

**4 Hard benchmarks remain unsaturated** HLE 44%, EnigmaEval 24%, and SciPredict 25% show substantial room for improvement.

**5 Muse Spark is a credible new entrant** Top-3 on 6+ benchmarks, including #1 on MultiChallenge at 75.52 and MCP Atlas at 78.30.

Evaluation takeaway: frontier capability is increasingly portfolio-shaped; model selection should map benchmark evidence to task domain rather than rely on a single overall winner.